

Inventory Management & Refund Portal — Functional Requirements

1. Purpose & Vision

Build a single internal portal to track and manage product inventory.

A key feature is a distinct, separate workflow for processing and managing product refunds, ensuring clarity and accuracy in stock adjustments.

2. Users & Roles

Administrator: Has full access to all system functionalities, including user management and system settings.

Staff: Has limited access, focused on daily operations, manages products and refunds.

Manager: Can approve/reject refund requests. Can see the dashboard

3. Data Model

This solution should have the following tables and attributes:

Customer

- Name
- Email
- Phone Number
- Billing Email
- Credit Limit
- Preferred Currency

Product

- SKU
- Product Name
- Product Description
- Price

- Available Stock
- Category_id (Product Category Identifier)
- Supplier_id (Supplier Identifier)

Product Category (Static)

- Gadgets
- Furniture
- Wellness

Supplier

- Supplier Name
- Contact Email
- Address
- Phone Number
- Website
- VAT Number
- Created Date

Refund

- Original_order_number
- Product_id
- Quantity Returned
- Return Reason
- Refund Status (Pending, Approved,/Rejected, Closed)
- Customer_Id
- Created_at
- Updated_at

4. Product Management

Admins must be able to add new products to the inventory. Required fields include: Product Name, SKU (Stock Keeping Unit), Category, Supplier, Cost Price, and Selling Price. An optional "Description" field should also be available.

All users must be able to view a paginated and searchable list of all products. The list should display key details like SKU, Product Name, Category, and Current Stock Level. Users should be able to sort and filter this list by any of its columns.

It should be possible to see product details when clicking on a specific product in the list.

Admins must be able to edit the details of any existing product (e.g., update price, change category).

Admins must be able to deactivate or soft-delete a product. A deactivated product will no longer appear in active searches or reports, but its historical data will be retained.

5. Product Refund Management Workflow

All users can initiate a refund request. This requires entering the original sales order number, the SKU of the product being returned, the quantity, and a reason for the refund (e.g., "Defective Item," "Customer Changed Mind"). The refund status should be defined as Draft.

An Admin must review each pending refund request. The Manager or Admin can either Approve or Reject the request. If rejected, the Admin must provide a reason.

Upon approval, the Manager or Admin should have a checkbox to confirm the customer has been financially refunded

All users must be able to view a complete history of all refund requests, filterable by status (Pending, Approved, Rejected), date, and SKU.

States: Draft → Pending → Approved or Rejected → Closed.

6. Reporting and Analytics

The system shall generate reports to provide insights into inventory and refund activities.

7. Theme

Use the following theme colours:

- Color-primary: #3D4E44;
- Color-secondary: #A69153;